

Abhinav Mahajan

abhinavm@andrew.cmu.edu | (412) 608-7868 | [abhinav-mahajan10.github.io](https://github.com/abhinav-mahajan10) | linkedin.com/in/abhinav-mahajan10

EDUCATION

Carnegie Mellon University

Master of Science in Computer Vision | GPA: 4.33/4.0

Pittsburgh, PA

December 2026

International Institute of Information Technology (IIIT Bangalore)

Bangalore, India

(Dual Degree) Master and Bachelor of Technology in Electronics and Communication Engineering July 2025

GPA: 3.63/4.00; Dean's Merit List; Teaching Assistant for Machine Learning (AI 511), Visual Recognition (AI 825)

PROFESSIONAL EXPERIENCE

Amazon

Bangalore, India

Applied Scientist Intern [Advisor: Arindam Sarkar, Dr. Prakash Mandayam Comar]

January 2025 - June 2025

- Achieved 3.2x top-K recall improvement in next-purchase prediction via MERIT, a novel exposure-mitigated training objective for autoregressive generative XMC with adaptive patching enabling 10x token compression.
- Distilled propensity-aligned embeddings from Phi-3.5 into a dual-encoder model, enabling bi-directional customers-categories ranking for personalized ad targeting at scale, achieving +0.26% conversion lift in A/B tests.
- Paper under review at KDD 2026; Received full-time offer for Applied Scientist-1 in the IML team, Bangalore.

Adobe Research

Bangalore, India

Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasani Srinivasan]

May 2024 - July 2024

- Developed Components-to-Design framework for harmonizing multimodal design assets (images, text, shapes) into aesthetic graphic designs (digital Posters/Flyers) using Large Multimodal Models and diffusion models.
- Designed a training-free identity-preserving composition method through cross-attention and self-attention matrix manipulation for LMM's, improving foreground identity retention while maintaining semantic coherence.
- US Patent no. 19/395,524; Paper under review at ICML 2026; Received full-time offer for MLE-1 in Firefly team.

Adobe Research

Bangalore, India

Research Intern [Advisor: Dr. KJ Joseph, Dr. Balaji Vasani Srinivasan]

May 2023 - August 2023

- Trained a self-supervised graphic design aesthetics scoring model using Siamese networks and contrastive learning to eliminate human annotation bias, achieving 94.97% accuracy in ranking graphic designs.
- Developed a novel genetic algorithm for layout refinement optimizing on the aesthetics scorer, outperforming all baselines; System deployed in Adobe Express; Paper accepted at WACV 2025 (Round 1, 13% acceptance rate).

RESEARCH EXPERIENCE

Research Intern - Human Sensing Lab, CMU [Advisor: Prof. Fernando De La Torre]

January 2026 - Present

- Developing compression algorithms of 4D Gaussian Splatting reconstructions for bandwidth-aware streaming.

Research Intern - Momentum Lab, CMU [Advisor: Prof. Jeff Ichnowski]

October 2025 - January 2026

- Developed two-stage pipeline: adaptation policy estimates rope parameters from safe exploratory motion ("wiggles"), informing trajectory model for goal-conditioned manipulation; Paper under review at RSS 2026.

MITACS Fellowship - Ontario Tech University [Advisor: Prof. Pooria Madani]

October 2025 - January 2026

- Awarded the MITACS Fellowship and interned at Ontario Tech University, Canada. Developed a platform to crowd source a dataset, contributed to research on screenshot parsing with VLMs and Active-Learning.

PUBLICATIONS

- [1] Jakobsson*, **Abhinav Mahajan*** et al. *Wiggle and Go! Deformable Object Rapid Motor Adaptation* **Under Review**
- [2] **Abhinav Mahajan*** et al. *MERIT: Mitigating Exposure Bias in Generative XMC for User Propensity*, **Under Review**
- [3] **Abhinav Mahajan*** et al. *Towards Design Generation from its Components*, **USPTO-19/395,524, Under Review**
- [4] Goyal*, **Abhinav Mahajan*** et al. *Design-o-meter: Towards Evaluating and Refining Graphic Designs*, **WACV 2025**
- [5] Ali Abedi*, Tracey Colella*, **Abhinav Mahajan** et al. *AVA: AI-driven Virtual Rehabilitation Assistant*, **WCISVR 2023**

SKILLS

Programming Languages/Tools: Advanced: Python, C++, Java, C, Javascript, Matlab, SQL, AWS, Docker

Libraries: Advanced: PyTorch, NumPy, Pandas, Diffusers, OpenCV, Transformers, trl, tensorflow

ML/CV Topics: Diffusion Models, LLMs, VLMs, Scene Understanding, Neural Radiance Fields, Gaussian Splatting

SELECTED PROJECTS

Flow Matching for 3D Shape Completion: Reformulated depth estimation as iterative refinement via learned velocity fields and LoRA adaptation for novel object categories, achieving 9.3% Chamfer Distance improvement on RaySt3R.

Pothole Detection and Severity Estimation: Won SDAIA Smartathon Hackathon (Road Safety Track) among 1,000+ submissions by training Mask R-CNN (82% accuracy) and 3D reconstruction pipeline for severity assessment.